Annals of Operations Research (2022) 312:853–881
https://doi.org/10.1007/s10479-022-04613-7

ORIGINAL RESEARCH

Portfolio optimization and marginal contribution to risk on multivariate normal tempered stable model

Young Shin Kim[1]

Accepted: 14 February 2022 / Published online: 9 March 2022
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract
This paper proposes a market model with returns assumed to follow a multivariate normal tempered stable distribution defined by a mixture of the multivariate normal distribution and the tempered stable subordinator. This distribution can capture two stylized facts: fat-tails and asymmetry, that have been empirically observed for asset return distributions. We discuss a new portfolio optimization method on the new market model, which is an extension of Markowitz's mean-variance optimization. The new optimization method considers not only reward and dispersion but also asymmetry in tails. The efficient frontier is extended to a curved surface on three-dimensional space of reward, dispersion, and asymmetry in tails. We also propose a new performance measure, which is an extension of the Sharpe ratio. Moreover, we derive closed-form solutions for portfolio managers' two important measures in portfolio construction: the marginal value-at-risk (VaR) and the marginal conditional VaR (CVaR). We illustrate the proposed model using stocks comprising the Dow Jones Industrial Average. First, perform the new portfolio optimization and then demonstrating how the marginal VaR and marginal CVaR can be used for portfolio optimization under the model. Based on this paper's empirical evidence, our framework offers realistic portfolio optimization and tractable methods for portfolio risk management.

Keywords Portfolio optimization · Asymmetry risk measure · Normal tempered stable distribution · Marginal contribution · Portfolio budgeting · Value at risk · Conditional value at risk

1 Introduction

It admits no doubt that the mean-variance model formulated by Markowitz (1952) is a major contribution to the portfolio theory in finance. Although some model assumptions were

The author gratefully acknowledges the support of GlimmAnalytics LLC and Juro Instruments Co., Ltd. The author is grateful to Minseob Kim, who reviewed this paper and corrected editorial errors. Also, all remaining errors are entirely my own.

✉ Young Shin Kim
aaron.kim@stonybrook.edu

[1] College of Business, Stony Brook University, New York, USA

challenged theoretically and empirically, it is impossible to exaggerate the importance of Markowitz's portfolio optimization theory. The optimization model has been applied to portfolio selection & allocation, and risk management.

This paper discusses two assumptions of the model and how we can improve the portfolio theory by relaxing the assumption of Markowitz's model. The first assumption is the use of the portfolio variance as a risk measure. Since asset return distribution does not follow Gaussian distribution but exhibits fat-tails and asymmetry, a risk measure would better be able to assess not only dispersion but also asymmetry in tails. Since Rubinstein (1973) and Kraus and Litzenberger (1976) discussed the skewness and higher moments in the portfolio allocation and valuation of risk assets, the idea of using skewness in portfolio optimization problems has been discussed in the literature. For example, Konno et al. (1993) add the skewness statistic directly into the portfolio optimization. Recently, Harvey (2010) and Adcock (2010) presented the portfolio optimization on the skewed normal distribution and Skewed-t distribution models, respectively. Portfolio optimization with asymmetry risk measures also has been discussed in King (1993) and Dahlquist et al. (2017). Since the asymmetric risk indirectly measured by the coherent risk measures such as the conditional value at risk (CVaR) by Pflug (2000) and Rockafellar and Uryasev (2000, 2002), the portfolio optimization with coherent risk measure has been studied in Rachev et al. (2007), Mansini et al. (2007), Rachev et al. (2011), and Kim et al. (2012). In this paper, the asymmetry tail risk is directly measured by a parametric method on a non-Gaussian asymmetric market model, which is discussed in the next paragraph.

The second is the assumption that asset returns follow a Gaussian distribution, which has been dominating financial theories. However, the Gaussian assumption was empirically rejected in literature including Mandelbrot (1963a, 1963b) and Fama (1963), and empirical evidence says that asset returns exhibit fat tails and asymmetry. Therefore, non-Gaussian models, which can describe the stylized facts about asset return data, have been suggested for the underlying model of the portfolio optimization. The subordinated Gaussian distribution is popularly used to construct a multivariate market model with fat-tails and asymmetry. This distribution of the model is defined by taking the multivariate normal distribution and changing the variance to a strictly positive random vector, which is referred to as the subordinator. Examples of the distribution used in the portfolio theory are α-stable subordinated Gaussian distribution (Rachev & Mittnik, 2000), the inverse Gaussian subordinated Gaussian distribution (Øigård et al., 2005; Aas et al., 2006; Eberlein & Madan, 2010), the inverse Gamma subordinated Gaussian distribution (Stoyanov et al., 2013) and the tempered stable subordinated Gaussian distribution (Barndorff-Nielsen & Shephard, 2001; Barndorff-Nielsen & Levendorskii, 2001).

This paper proposes a non-Gaussian market model that returns are assumed to follow the normal tempered stable (NTS) distribution. The NTS distribution is the tempered stable subordinated Gaussian distribution. It is asymmetric and has exponential tails that are fatter than Gaussian tails and thinner than the power tails of α-stable distributions. For that reason, it can describe the asymmetric and fat-tail properties of the stock return distribution. Since it has exponential tails, it has finite exponential moments and finite integer moments for all orders, such as mean, variance, skewness, and kurtosis. By standardization, we obtain a NTS distribution having zero mean and unit variance. That can be used as the innovation distribution of time series models, including the ARMA–GARCH model. Since the distribution is infinitely divisible, we can generate a continuous-time Lévy process on the NTS distribution. For this good properties, the distribution was popularly used in finance; portfolio optimization (Eberlein & Madan, 2010; Kim et al., 2012; Anand et al., 2016), risk management (Kim et al., 2010; Anand et al., 2017; Kurosaki & Kim, 2018), option pricing (Boyarchenko & Lev-

endorskiĭ, 2000; Rachev et al., 2011; Eberlein & Glau, 2014; Kim et al., 2015), term structure of interest rate model (Eberlein & Özkan, 2005), and credit risk management (Eberlein et al., 2012; Kim & Kim, 2018). Recently, the expectation-maximization maximum likelihood (EM-MLE) algorithm for the multivariate NTS model was studied in Bianchi et al. (2016) and Bianchi and Tassinari (2020). The multivariate model also applied to the option pricing in Kim et al. (2015), and Fallahgoul et al. (2019). Bianchi and Tassinari (2020) discussed the risk-neutral measure on the multivariate NTS model and its application to the option pricing and portfolio optimization. Moreover, Xia and Grabchak (2021) presented estimation and simulation method of the model with finance data, and Kim and Volkmann (2013) and Liu et al. (2021) applied the model to the systemic risk analysis using the NTS copula.

The subordinated Gaussian distribution can be extended to the subordinated infinitely divisible distributions or the subordinated Lévy process. For example, the stochastic volatility Lévy process model by Carr et al. (2003) used the CIR process as the subordinator of the CGMY.[1] Lévy process and found the risk-neutral price of options. Moreover, Hitaj and Mercuri (2019) defined a new multivariate distribution using Gamma subordinated CGMY distribution and applied it to the portfolio optimization. Different from Hitaj et al. (2019), we decompose the market risk to dispersion risk and asymmetric tail risk in this paper and discuss portfolio optimization in view of those two risk values on the NTS distribution model.

An important measure derived from portfolio optimization is the marginal risk contribution. The marginal risk contribution help managers to make portfolio rebalancing decisions or re-optimization decisions. This risk measure is the rate of change in risk, whether it is variance, value-at-risk (VaR), or CVaR, with respect to a small percentage change in the size of a portfolio allocation weight. Mathematically it is defined by the first derivative of the risk measure with respect to the marginal weight. Because of the importance of this measure in portfolio decisions, a closed-form solution for this measure is needed. The general form of marginal risk contributions for the VaR and CVaR are provided in Gourieroux et al. (2000). Moreover, the closed-form of the marginal risk contributions for VaR and CVaR are discussed under the skewed-t distributed market model in Stoyanov et al. (2013), and under the NTS market model in Kim et al. (2012).

The contributions of this paper are as follows. First, we construct a market model using the multivariate NTS distributed portfolio returns. Unlike the NTS market model in Kim et al. (2012), the market model in this paper is constructed by only the standard NTS distribution, a subclass of NTS distribution having zero mean and unit variance. Using the new NTS market model, we discuss the portfolio optimization theory considering dispersion risk and asymmetric tail risk. As the Markowitz model, the standard deviation is used for the dispersion risk measure. In addition to the dispersion risk measure, the asymmetric tail risk measure is proposed in order to capture the asymmetry directly. It is a weighted mean of asymmetry parameters of the standard NTS distribution. Using those two risk measures, we find an efficient frontier surface extending Markowitz's efficient frontier. Moreover, a new performance measure of a portfolio on an efficient frontier surface is presented. The new performance measure is an extension of the Sharpe ratio (Sharpe, 1966, 1994). Finally, we provide closed-form solutions of the marginal risk contribution for VaR and CVaR under the NTS market model. The marginal VaR and marginal CVaR formula in this paper are simpler than the solutions presented in Kim et al. (2012), and hence we can discuss the iterative risk budgeting, which was not discussed in Kim et al. (2012).

Empirical illustrations are provided for each topic in the paper with performance tests. Data used in the empirical illustrations are historical daily returns of major 30 stocks in the

[1] See Carr et al. (2002), Koponen (1995), and Boyarchenko and Levendorskiĭ (2000)

U.S. market. We draw the efficient frontier surface based on the estimated parameters of the NTS market model. The performance measure maximization strategy is also exhibited and it is compared to the traditional Sharpe ratio maximization strategy by backtesting. We calculate Marginal VaR & CVaR and perform the risk budgeting using calculated marginal VaR & CVaR.

The remainder of this paper is organized as follows. The NTS market model constructed on the multivariate standard NTS distributed noise is presented in Sect. 2. In this section, we show that the parameters of the portfolio return distribution are estimated by marginal parameters and the covariance matrix of individual stock returns instead of the covariance matrix of the standard NTS noises. The portfolio optimization with dispersion and asymmetric tail risk measures is discussed in Sect. 3 under the NTS market model. The new performance measure of the portfolio is presented in Sect. 4. We provide closed-form solutions for the marginal VaR and CVaR under the NTS market model in Sect. 5. We also discuss portfolio budgeting using the marginal VaR and the marginal CVaR. Finally, Sect. 6 concludes. Proofs and mathematical details are presented in the "Appendix".

2 Normal tempered stable market model

Let N be a finite positive integer and $X = (X_1, X_2, \cdots, X_N)^{\mathrm{T}}$ be a multivariate random variable given by

$$X = \beta(\mathcal{T} - 1) + \mathrm{diag}(\gamma)\varepsilon\sqrt{\mathcal{T}},$$

where

- $\mathcal{T}$ is the tempered stable subordinator.[2] with parameters (α, θ), and is independent of ε_n for all $n = 1, 2, \ldots, N$.
- $\beta = (\beta_1, \beta_2, \cdots, \beta_N)^{\mathrm{T}} \in \mathbb{R}^N$ with $|\beta_n| < \sqrt{\frac{2\theta}{2-\alpha}}$ for all $n \in \{1, 2, \ldots, N\}$.
- $\gamma = (\gamma_1, \gamma_2, \cdots, \gamma_N)^{\mathrm{T}} \in \mathbb{R}_+^N$ with $\gamma_n = \sqrt{1 - \beta_n^2 \left(\frac{2-\alpha}{2\theta}\right)}$ for all $n \in \{1, 2, \ldots, N\}$ and $\mathbb{R}_+ = [0, \infty)$.
- $\varepsilon = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_N)^{\mathrm{T}}$ is N-dimensional standard normal distribution with a covariance matrix Σ. That is, $\varepsilon_n \sim \Phi(0, 1)$ for $n \in \{1, 2, \ldots, N\}$ and (k, l)th element of Σ is given by $\rho_{k,l} = \mathrm{cov}(\varepsilon_k, \varepsilon_l)$ for $k, l \in \{1, 2, \ldots, N\}$. Note that $\rho_{k,k} = 1$.

In this case, X is referred to as the *N-dimensional standard NTS random variable* with parameters $(\alpha, \theta, \beta, \Sigma)$ and we denote it by $X \sim \mathrm{stdNTS}_N(\alpha, \theta, \beta, \Sigma)$ (See the "Appendix").

Consider a portfolio having N assets. The return of the assets in the portfolio is given by a random vector $R = (R_1, R_2, \cdots, R_N)^{\mathrm{T}}$. We suppose that the return R follows

$$R = \mu + \mathrm{diag}(\sigma)X \tag{1}$$

where $\mu = (\mu_1, \mu_2, \cdots, \mu_N)^{\mathrm{T}} \in \mathbb{R}^N$, $\sigma = (\sigma_1, \sigma_2, \cdots, \sigma_N)^{\mathrm{T}} \in \mathbb{R}_+^N$ and $X \sim \mathrm{stdNTS}_N(\alpha, \theta, \beta, \Sigma)$. Then we have $E[R_n] = \mu_n$ and $\mathrm{var}(R_n) = \sigma_n^2$ for all $n \in \{1, 2, \cdots, N\}$. This market model is referred to as the *NTS market model*. Let $w = (w_1, w_2, \cdots, w_N)^{\mathrm{T}} \in I^N$ with $I = [0, 1]$ be the capital allocation weight vector.[3] Then the portfolio return for w is equal to $R_P(w) = w^{\mathrm{T}}R$. The distribution of $R_P(w)$ is presented in the following proposition whose proof is in "Appendix".

[2] The tempered subordinator is defined by the characteristic function (12) in the "Appendix"

[3] In this paper, we consider the long only portfolio.

Table 1 Companies and symbols of 30 Stocks

Company	Symbol	Company	Symbol
3M	MMM	American Express	AXP
Apple Inc.	AAPL	Boeing	BA
Caterpillar Inc.	CAT	Chevron Corporation	CVX
Cisco Systems	CSCO	The Coca-Cola Company	KO
DuPont de Nemours Inc.	DD	Exxon Mobil	XOM
Goldman Sachs	GS	The Home Depot	HD
IBM	IBM	Intel	INTC
Johnson & Johnson	JNJ	JPMorgan Chase	JPM
McDonald's	MCD	Merck & Co.	MRK
Microsoft	MSFT	Nike	NKE
Pfizer	PFE	Procter & Gamble	PG
Raytheon Technologies	RTX	The Travelers Companies	TRV
United Health Group	UNH	Verizon	VZ
Visa Inc.	V	Walmart	WMT
Walgreens Boots Alliance	WBA	The Walt Disney Company	DIS

They are selected based on the components for DJIA index, since April 6, 2020, but Dow Inc. (DOW) in the components is replaced by DuPont de Nemours Inc. (DD)

Proposition 1 *Let* $\mu \in \mathbb{R}^N$, $\sigma \in \mathbb{R}^N_+$, *and* $X \sim \text{stdNTS}_N(\alpha, \theta, \beta, \Sigma)$. *Suppose a* N*-dimensional random variable* R *is given by* (1) *and* $w \in I^N$ *with* $I = [0, 1]$. *Then*

$$R_P(w) \overset{\text{d}}{=} \bar{\mu}(w) + \bar{\sigma}(w)\Xi \quad \textit{for} \quad \Xi \sim \text{stdNTS}_1(\alpha, \theta, \bar{\beta}(w), 1), \tag{2}$$

where

$$\bar{\mu}(w) = w^T\mu, \quad \bar{\sigma}(w) = \sqrt{w^T\Sigma_R w}, \quad \bar{\beta}(w) = \frac{w^T\text{diag}(\sigma)\beta}{\bar{\sigma}(w)}$$

and Σ_R *is the covariance matrix of* R.

According to Proposition 1, we need only Σ_R, which is covariance matrix of R, and we do not need to know Σ, which is the covariance matrix for ϵ, when we study the portfolio return of the NTS market model.

Empirical Illustration

We fit the NTS market model to 30 major stocks in the U.S. stock market. The 30 stocks are selected based on the components for Dow Johns Industrial Average (DJIA) index, since April 6, 2020. However, Dow Inc.(DOW) in the components is replaced by DuPont de Nemours Inc.(DD), since DOW does not have enough history. Table 1 exhibits the names of companies and symbols of those 30 Stocks. We calculate the sample mean and sample standard deviation of daily log-returns for each stock and then fit the stdNTS parameters using standardized residual. Since the subordinator process is not directly observed in the financial data, it is an important problem to estimate α and θ, which are the subordination parameters of the NTS distribution. Kim and Volkmann (2013) and Kim et al. (2015) used the maximum likelihood estimation (MLE) directly to multivariate data (2-dimensional data) and found α, θ, and a vector of β. However, MLE is required long computation time and extremly large memory space, and it can be used only for the bivariate NTS model. Since

we must fit 30-dimensional distribution in this case, the MLE cannot be practically applied. For that reason, the index-based method has been used in the literature, including Kim et al. (2012), and Anand et al. (2016, 2017). [4]

In this parameter fitting, the curve fit method is used between the cumulative distribution function (CDF) of the stdNTS distribution.[5] and the empirical CDF obtained by the kernel density estimation. In order to find α and θ, we use the index-based method with DJIA index data. That is we find the 30-dimensional stdNTS parameters as the following two-step method:

Step 1 Find $(\alpha_{DJ}, \theta_{DJ}, \beta_{DJ})$ using the curve fit method between the empirical CDF and stdNTS CDF for the standardized return data of DJIA index.
Step 2 Fix $\alpha = \alpha_{DJ}$ and $\theta = \theta_{DJ}$. Find β_n using the curve fit method again for each nth stock returns $n \in \{1, 2, \ldots, 30\}$ with fixed α and θ.

The estimated parameters of μ_n, σ_n and β_n are presented in Table 2. The fixed α and θ are in the bottom of the table. Those parameters are estimated using 1-day log-returns of the 30 stocks from 1/1/2017 to 12/31/2019.[6] We can see that only 5 stocks (BA, DD, DIS, JPM, NKE) out of those 30 stocks have positive betas in this table. That means, 25 stocks follow left-skewed return distributions, while only 5 stocks follow positively skewed distributions. The parameter α closes to 1, and θ is less than 1. That means the 30-dimensional distribution for those 30 stock returns has fat-tails.[7] We perform the Kolmogorov–Smirnov (KS) goodness of fit test. KS statistic (KS) values are presented in the table. We do not need to estimate the covariance matrix Σ for ε, but to estimate the covariance matrix Σ_R of R as discussed in Proposition 1.

3 Portfolio optimization and efficient frontier

In traditional mean-variance portfolio optimization, the investor finds the optimal portfolio which maximizes the reward and minimizes the dispersion risk. The reward is the expected return of the portfolio, and the dispersion risk is the standard deviation (or the variance) of the portfolio. However, the dispersion risk is only one feature of the portfolio risk, while there are many other risks, for instance, the asymmetric tail risk. If there are two portfolios having same mean and variance, and one of them follows a skewed right distribution and the other follows a skewed left distribution, then the skewed left distributed portfolio is riskier than the skewed right distributed portfolio. We take the NTS market model and measure both the dispersion risk and asymmetric tail risk of a given portfolio.

Suppose that the marginal return of the portfolio is given by the vector R in (1), then the portfolio return $R_p(w)$ for the given portfolio allocation vector w is equal to (2). The classical dispersion risk measure introduced by Markowitz (1952) is the standard deviation of the portfolio defined as follows:

$$\text{Disp.}(w) = \bar{\sigma}(w) = \sqrt{w^{\mathrm{T}} \Sigma_R w},$$

[4] Kim et al. (2012) and Anand et al. (2016) used the DJIA index, and Anand et al. (2017) used the S&P 1200 financial index (SGFS) as the index to estimate the parameters (α, θ).

[5] CDF of stdNTS distribution is obtained by the fast Fourier transform method by Gils-Pelaez (1951).

[6] We select the sample size 752 (3 years) to see at least two outliers (left and right tails). If the data is normal, we expect more than 2 outliers which are not included 3σ range covers 99.7% out of 752 samples.

[7] More precisely, it has the semi-fat-tails having the exponential decaying tails

Table 2 NTS parameter fit using 1-day-returns from 1/1/2017 to 12/31/2019

Symbol	$\mu_n(\%)$	$\sigma_n(\%)$	$\beta_n(\%)$	KS
AAPL	0.13	1.56	− 2.55	0.022
AXP	0.073	1.18	− 3.90	0.036
BA	0.11	1.68	0.50	0.038
CAT	0.071	1.71	−2.93	0.035
CSCO	0.071	1.44	−8.56	0.015
CVX	0.018	1.22	−4.65	0.032
DD	− 0.020	1.70	0.32	0.034
DIS	0.046	1.24	2.00	0.029
GS	− 0.00044	1.46	− 1.01	0.043
HD	0.074	1.16	− 7.01	0.044
IBM	− 0.013	1.30	− 7.31	0.012
INTC	0.075	1.70	− 1.19	0.024
JNJ	0.041	1.09	− 7.53	0.020
JPM	0.072	1.21	3.85	0.042
KO	0.050	0.89	− 2.67	0.032
MCD	0.076	1.04	− 3.17	0.024
MMM	0.0093	1.36	− 10.55	0.0084
MRK	0.067	1.14	− 0.68	0.037
MSFT	0.13	1.36	− 4.47	0.027
NKE	0.093	1.52	0.44	0.030
PFE	0.037	1.08	− 7.19	0.032
PG	0.064	1.00	− 0.26	0.025
RTX	0.050	1.22	− 7.41	0.024
TRV	0.025	1.07	− 10.52	0.036
UNH	0.086	1.35	− 1.41	0.029
V	0.12	1.22	− 10.07	0.023
VZ	0.033	1.12	− 4.12	0.040
WBA	− 0.036	1.58	− 12.62	0.021
WMT	0.083	1.21	− 1.64	0.019
XOM	− 0.019	1.12	− 4.16	0.051*

$\alpha = 0.9766$ and $\theta = 0.2253$

* p value for the KS statistic of XOM is 4.14% that is less than 5%

where w is the capital allocation weight of the portfolio. The asymmetric parameter $\bar{\beta}(w) = \frac{w^T \mathrm{diag}(\sigma)\beta}{\bar{\sigma}(w)}$. After extracting the dispersion effect $\mathrm{diag}(\sigma)/\bar{\sigma}(w)$ from $\bar{\beta}(w)$, we define the asymmetric tail risk as follows:

$$\mathrm{TailRisk}(w) = w^{\mathrm{T}}\beta,$$

which is related to only asymmetric tails. Since $|\beta_n| < \sqrt{\frac{2\theta}{2-\alpha}}$ for all $n \in \{1, 2, \ldots, N\}$ by the definition of the NTS market model, $|\mathrm{TailRisk}(w)|$ cannot be larger than $\sqrt{\frac{2\theta}{2-\alpha}}$, if $0 \leq w_n \leq 1$ for all $n \in \{1, 2, \ldots, N\}$. A portfolio having positive $\mathrm{TailRisk}(w)$ follows the right-skewed distribution, and a portfolio having negative $\mathrm{TailRisk}(w)$ follows the left-skewed distribution. If there are two portfolios, A and B, whose capital allocation weight

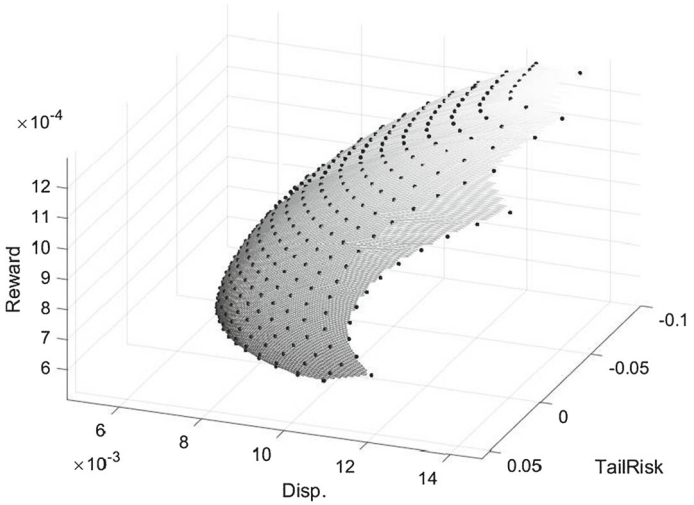


Fig. 1 Surface of efficient frontiers

vectors are w_A and w_B respectively, and $\text{TailRisk}(w_A) < \text{TailRisk}(w_B)$, then portfolio A is riskier than portfolio B if they have the same means the same variances. More precisely, VaR and CVaR of portfolio A is larger than those values of portfolio B. As we discuss in the introduction, the skewness and higher moments has been used in portfolio optimization in the literature. However, higher moments of the portfolio return $R_p(w)$ are formulated to nonlinear functions for w, and that makes the portfolio optimization inefficient. While, the formula of $\text{TailRisk}(w)$ is linear and it effect on the higher moments of the portfolio return (See Remark 1 in the "Appendix").

Based on those two risk measures, we set a nonlinear programming problem for the portfolio optimization as

$$\begin{aligned} &\min_{w} \ (\text{Disp.}(w)) \\ \text{subject to} \ &\sum_{n=1}^{N} w_n = 1 \\ &w_n \geq 0 \text{ for all } n \in \{1, 2, \ldots, N\} \\ &\text{TailRisk}(w) = w^{\mathrm{T}}\beta \geq b^* \\ &\text{Reward}(w) = w^{\mathrm{T}}\mu \geq m^* \end{aligned}$$

where the benchmark values for Portfolio reward and asymmetric tail risk are $m^* \in [\min(\mu), \max(\mu)]$ and $b^* \in [\min(\beta), \max(\beta)]$, respectively.

Using the parameters in Table 2, we perform the portfolio optimization for 51 points of b^* in $\{b = \min(\beta) + k \cdot (\max(\beta) - \min(\beta))/50 \,|\, k = 0, 1, 2, \ldots, 50\}$ and for 51 points of m^* in $\{m = \min(\mu) + k \cdot (\max(\mu) - \min(\mu))/50 \,|\, k = 0, 1, 2, \ldots, 50\}$. We finally obtain the efficient frontier surface on the three-dimensional space in Fig. 1. The dot-points of the surface is $(\text{Disp.}(w), \text{TailRisk}(w), \text{Reward}(w))$ for the optimal capital allocation weight w. The mash surface of the figure is an interpolation surface of the dot-points.

In order to look into the surface, Fig. 2 provides three examples of efficient frontier curves between the dispersion risk and the reward. The three curves are extracted from Fig. 1. For

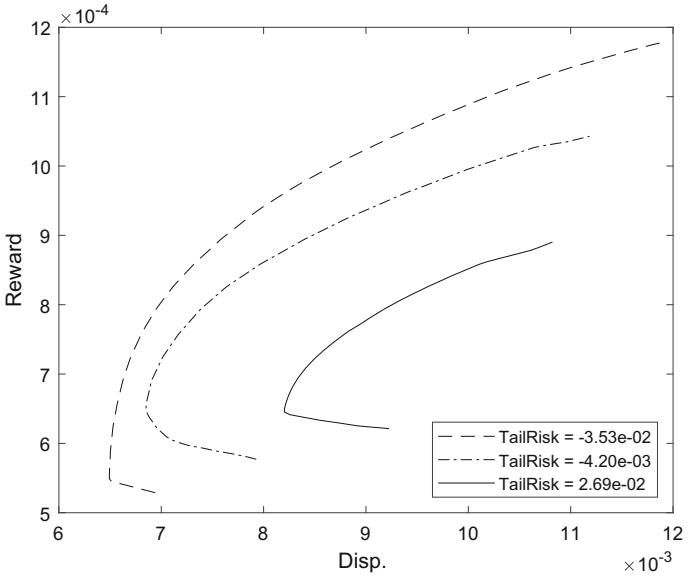


Fig. 2 Efficient frontiers between the dispersion risk and the reward. We can observe that the mean-variance efficient frontier is not unique but changes with respect to the asymmetric tail risk

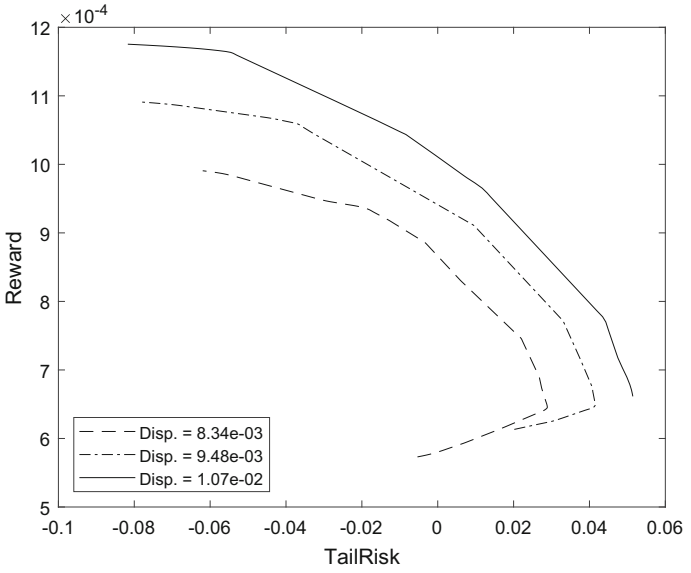


Fig. 3 Efficient frontiers between the asymmetric tail risk and the reward. We can observe that the reward is decreasing when the asymmetric tail risk is increasing under the fixed dispersion risk

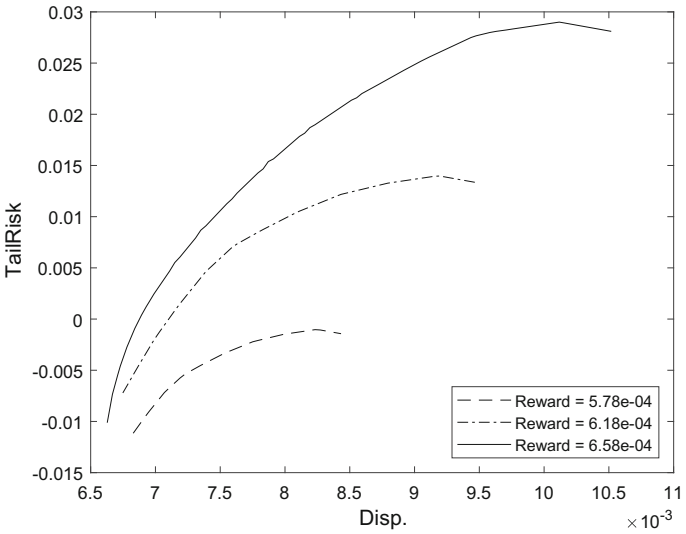


Fig. 4 Efficient frontiers between the dispersion risk and the asymmetric tail risk. We can observe that the asymmetric tail risk is decreasing when the dispersion risk is decreasing under the fixed reward

instance, we fix $b^* = -1.64\%$, and perform the optimization for $m^* \in [\min(\mu), \max(\mu)] = [-0.036\%, 0.13\%]$, present it to the dash curve of the figure. Using the same method, we draw the dash-dot curve and the solid curve for $b^* = 0.187\%$ and $b^* = 2.02\%$, respectively. Figure 3 presents another examples of efficient frontier curves between the asymmetric tail risk and the reward. Those three curves are extracted from Fig. 1, too. The solid curve is for (TailRisk(w), Reward(w)) having Disp.$(w) = 0.840\%$. Also, the dash-dot and the dashed curves are for Disp.$(w) = 0.961\%$ and for Disp.$(w) = 1.09\%$, respectively. Since the constraint TailRisk$(w) = w^{\mathrm{T}}\beta \geq b^*$ is linear, the efficient frontiers have polygon shape. Figure 4 present the efficient frontier curves between the dispersion risk and the asymmetric tail risk extracted from Fig. 1. The solid, dash-dot, and dashed curves are for (Disp.(w), TailRisk(w)) with Reward$(w) = 0.578\%$, Reward$(w) = 0.618\%$, and Reward$(w) = 0.658\%$, respectively. According to those three figures, we observe that the classical (mean-variance) efficient frontier is not unique but there are many various form of efficient frontier curves with respect to the asymmetric tail risk. Moreover, we can observe that the reward is decreasing when the asymmetric tail risk is increasing under the fixed dispersion risk.

4 Performance measure

According to Fig. 4, we can see that the smaller Disp.(w) the smaller TailRisk(w) if Reward(w) value is fixed. Since the portfolio having less TailRisk(w) value is riskier, the portfolio with less Disp.(w) is not always less risky. The portfolio with smaller Disp.(w) and larger TailRisk(w) performs better if Reward(w) value is fixed in this model. The Sharpe ratio (Sharpe, 1966, 1994) considers only the dispersion risk to measure the performance of the portfolio. In this section, we present a new performance measure on the NTS Market model. The new performance measure is an extension of the Sharpe ratio, and it measures

the performance of portfolios considering not only the dispersion risk but also asymmetric tail risk.

The Sharpe ratio is defined as

$$\mathcal{R}_S(w, r_f) = \frac{\text{Reward}(w) - r_f}{\text{Disp.}(w)}$$

where r_f is the risk free rate of return, and the tangency portfolio is the Sharpe ratio maximizing portfolio on the efficient frontier. In the NTS market model, the efficient frontier is not given by unique curve but surface, we have many tangency portfolios with respect to asymmetric tail risks. For each TailRisk(w), we define the tangency portfolio as

$$w_{tangency}(b) = \arg\max_{w \in \mathcal{S}} \left\{ \mathcal{R}_S(w, r_f) \middle| \text{TailRisk}(w) \geq b \right\},$$

where $\mathcal{S} = \{w = (w_1, \ldots, w_N) \in I^N \mid I = [0, 1], \sum_{n=1}^N w_n = 1\}$.

We define a new performance measure as

$$\mathcal{R}_{AS}(b, r_f) = \frac{\mathcal{R}_S(w_{tangency}(b), r_f)}{A(b)},$$

where

$$A(b) = \frac{\sqrt{\frac{2\theta}{2-\alpha}} - b}{2\sqrt{\frac{2\theta}{2-\alpha}}}$$

for the level b of asymmetry tail risk. The measure $\mathcal{R}_{AS}(b, r_f)$ is referred to as the *asymmetry scored Sharpe ratio* or simply *AS ratio*. The value $A(b)$ increases when b decreases, and $A(b)$ decreases when b increases. Moreover, $0 \leq A(\text{TailRisk}(w)) \leq 1$ since $|\text{TailRisk}(w)| \leq \sqrt{\frac{2\theta}{2-\alpha}}$ if $w \in I^N$ with $I = [0, 1]$. Hence, we can use $A(\text{TailRisk}(w))$ as the evaluation index for a portfolio. Suppose we have two portfolios P_w and P_v having capital allocation weights w and v, respectively. Assume that those two portfolio have the same Sharpe ratio. If $\text{TailRisk}(w) > \text{TailRisk}(v)$ i.e. $A(\text{TailRisk}(w)) < A(\text{TailRisk}(v))$ then a risk averse investor may select the portfolio P_w instead of P_v because the return distribution of P_v has the fatter negative tail than the return distribution of P_w.

Using the AS ratio, Sharpe ratio of the tendency portfolio is evaluated by the level of the asymmetric tail risk. We find the AS ratio maximizing portfolio as follows:

$$P^* = \max_{b \in [\min(\beta), \max(\beta)]} \mathcal{R}_{AS}(b, r_f)$$

The b^* corresponding to the b value for the AS ratio maximizing portfolio P^* is

$$b^* = \arg\max_{b \in [\min(\beta), \max(\beta)]} \mathcal{R}_{AS}(b, r_f).$$

The capital allocation weight vector w^* corresponding to P^* is $w^* = w_{tangency}(b^*)$.

Figure 5 presents the AS ratio curve with respect to the asymmetric tail risk based on $r_f = 0.025/252$ and the estimated market parameters in Table 2. The solid curve is a $\mathcal{R}_{AS}(b, r_f)$ for $b \in [\min(\beta), \max(\beta)]$, and the dot point is the AS ratio maximizing portfolio $(b^*, P^*) = (-0.0187, 0.2299)$.

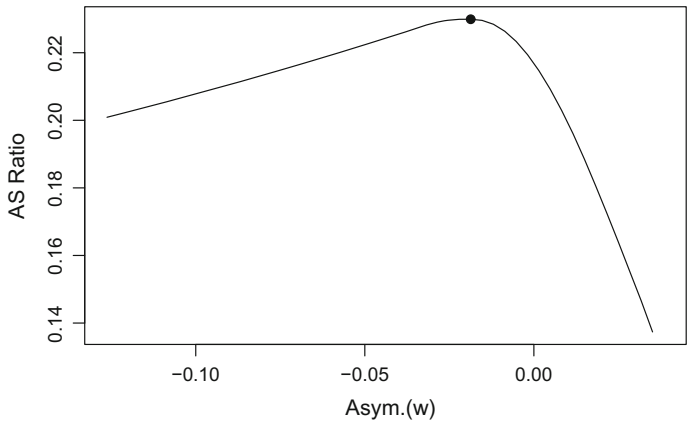


Fig. 5 AS ratio curve. The x-axis is for the level b of asymmetric tail risk and the y-axis is for AS ratios

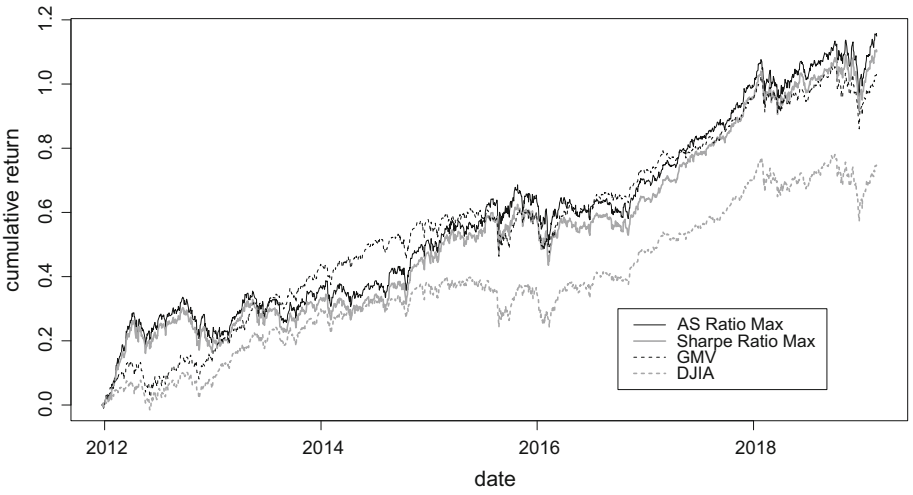


Fig. 6 Cumulative return curves for the AS ratio maximization strategy (solid black), the Sharpe ratio maximization strategy (solid gray), GMV portfolio (dashed black), and DJIA index (dashed gray)

Trading strategies

We estimate parameters and rebalance portfolio every 10 days from 12/23/2011 to 02/21/2019. We perform the parameter estimation explained in Sect. 2, and then find the AS ratio maximizing portfolio, the Sharpe ratio maximizing portfolio and global minimum variance (GMV) portfolio.[8] every 10 days with constraints that $w_n \geq 0$ for all $n \in \{1, 2, \ldots, 30\}$ and $\sum_{n=1}^{30} w_n = 1$. We set the risk free rate of return to be zero to simplify this investigation. We do not change those three portfolios for 10 days and re-balance them on the first day of the next 10 days. Figure 6 provides daily cumulative returns for those three strategies (the AS ratio maximizing, GMV, and the Sharpe ratio maximizing strategies). As a benchmark, we add the DJIA index. In Table 3, we can see the performance of this investigation. The stdNTS parameters are also provided in the table for those four cases.

[8] See Stoyanov et al. (2010) and Hitaj et al. (2019).

Table 3 Performance of the AS ratio maximization strategy, the Sharpe ratio maximization strategy, GMV portfolio, and DJIA index

	AS ratio max	Sharpe ratio max	GMV	DJIA
Mean	0.0645%	0.0618%	0.0576%	0.0419%
Std	0.984%	0.942%	0.830%	0.804%
α	0.46	0.56	0.75	0.39
θ	1.09	0.96	0.40	0.79
β	− 0.0603	− 0.0712	− 0.0262	− 0.0525
$\text{VaR}_{1\%}$	2.49%	2.42%	2.32%	2.34%
$\text{CVaR}_{1\%}$	3.22%	3.14%	3.07%	3.01%
Max. Drawdown	18.55%	18.98%	19.39%	20.79%
Sharpe ratio	6.61%	6.56%	6.94%	5.21%
VaR ratio	2.59%	2.56%	2.49%	1.79%
CVaR ratio	2.01%	1.97%	1.88%	1.39%
$\mathcal{R}_{AS}(\beta, r_f)$	12.57%	12.35%	13.45%	9.89%

std dev standard deviation

The AS ratio maximization strategy has the largest mean return, and largest VaR and CVaR ratios in this investigation. We can say that the AS ratio maximization strategy takes higher risks than the other three cases, and hence it obtains a higher return. Moreover, the maximum drawdown of the AS ratio maximization strategy is less than in other cases. This investigation is an out-of-sample test, the AS ratio maximization strategy is not always the best. GMV portfolio has the smallest standard deviation, VaR and CVaR, in this investigation. Sharpe ratio of GMV portfolio is even larger than the value of Sharpe ratio maximization strategy. The $\mathcal{R}_{AS}(\beta, r_f)$ value of the AS ratio maximization strategy is larger than the values of the Sharpe ratio maximization strategy and DJIA index, but smaller than the values of the GMV portfolio.

To discuss the market concentration of the portfolio, the Herfindahl–Hirschman index (HHI) curves for the AS ratio maximization and Sharpe ratio maximization strategies are exhibited in Fig. 7. At each re-balancing date, we calculate HHI for the portfolio weight vector as follows

$$HHI = \sum_{n=1}^{30} w_n^2$$

and draw them as a timeseries curve in the figure. Based on the graph, both the AS ratio maximization and the Sharpe ratio maximization strategies have high concentration level comparing to the GMV portfolio. The concentration level of the AS ratio maximization is relatively higher than the Sharpe ratio maximization strategy, but the difference is not remarkable.

5 Marginal contribution to risk

In this section, we discuss the marginal contribution to risk on the NTS market model. We consider VaR and CVaR as risk measures, and find analytic solutions of the marginal contribution to VaR and to CVaR, respectively. When portfolio managers make portfolio

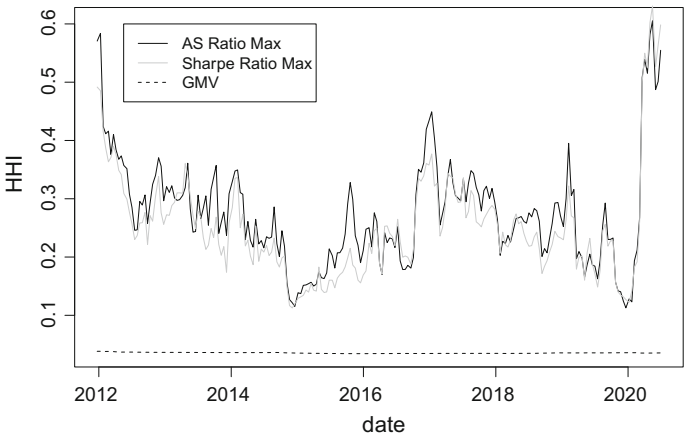


Fig. 7 HHI timeseries for the AS ratio maximization strategy (solid black), the Sharpe ratio maximization strategy (solid gray) and GMV portfolio (dashed black)

rebalancing decisions or re-optimization decisions, the marginal contribution to risk is used as an important risk measure derived from portfolio optimization. This risk measure can be thought of as the rate of change in risk with respect to a small percentage change in the size of a portfolio holding weight. That is, the measure can be obtained by the first derivative of the risk measure with respect to the holding's marginal weight. Since the portfolio rebalancing decision must be made quickly in the intra-day trading, we need a closed-form solution for the marginal contribution to risk.

Suppose R is portfolio return as (1) in the NTS market model. Let w be a capital allocation rate vector, then $R_P(w) = w^{\mathrm{T}}R$ is given by (2). Let $F_{\mathrm{stdNTS}}(x, \alpha, \theta, \beta)$, $F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta)$ and $\phi_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta)$ be the CDF, the inverse CDF, and the characteristic function (Ch.F) of $\mathrm{stdNTS}_1(\alpha, \theta, \beta, 1)$. Then we have

$$\mathrm{VaR}_\eta(\Xi) = -F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w)), \quad \text{and} \quad \phi_\Xi(-u + i\delta) = \phi_{\mathrm{stdNTS}}(-u + i\delta, \alpha, \theta, \bar{\beta}(w)).$$

Moreover, let $\mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)$ be the CVaR value of $\mathrm{stdNTS}_1(\alpha, \theta, \beta, 1)$ at the tail level η. If there is $\delta > 0$ such that $|\phi_\Xi(-u + i\delta)| < \infty$ for all $u \in \mathbb{R}$ then we have.[9]

$$\begin{aligned}&\mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)\\ &= -F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta) - \frac{1}{\pi\eta}\mathrm{Re}\int_0^\infty e^{(iu+\delta)F^{-1}_{\mathrm{stdNTS}}(u,\alpha,\theta,\beta)}\frac{\phi_{\mathrm{stdNTS}}(-u + i\delta, \alpha, \theta, \beta)}{(-u + i\delta)^2}du,\end{aligned} \tag{3}$$

and

$$\mathrm{CVaR}_\eta(\Xi) = \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w)).$$

Therefore, the VaR and CVaR of the portfolio return $R_P(w)$ are as follows:

$$\begin{aligned}\mathrm{VaR}_\eta(R_p(w)) &= -\bar{\mu}(w) + \bar{\sigma}(w)\mathrm{VaR}_\eta(\Xi)\\ &= -\bar{\mu}(w) - \bar{\sigma}(w)F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w))\end{aligned}$$

[9] See Proposition 2 in Kim et al. (2010) and (2012).

and

$$\begin{aligned}\mathrm{CVaR}_\eta(R_p(w)) &= -\bar{\mu}(w) + \bar{\sigma}(w)\mathrm{CVaR}_\eta(\Xi)\\ &= -\bar{\mu}(w) + \bar{\sigma}(w)\mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w)).\end{aligned}$$

The first derivative of $\mathrm{VaR}_\eta(R_P(w))$ and the first derivative of $\mathrm{CVaR}_\eta(R_P(w))$ for w_n are referred to as *the marginal contribution to VaR* (MCT-VaR) and *the marginal contribution to CVaR* (MCT-CVaR), respectively. They are provided the following proposition whose proof is in the "Appendix".

Proposition 2 *Suppose a portfolio return R follows the NTS market model as (1). Let w be a capital allocation rate vector, and $R_P(w) = w^T R$. The marginal contribution to VaR and to CVaR are equal to*

$$\begin{aligned}\frac{\partial \mathrm{VaR}_\eta(R_p(w))}{\partial w_n} &= -\mu_n - F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w))\frac{\partial \bar{\sigma}(w)}{\partial w_n}\\ &\quad - \left(\sigma_n \beta_n - \bar{\beta}(w)\frac{\partial \bar{\sigma}(w)}{\partial w_n}\right)\frac{\partial F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta}\Big|_{\beta=\bar{\beta}(w)} \qquad (4)\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \mathrm{CVaR}_\eta(R_p(w))}{\partial w_n} &= -\mu_n + \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w))\frac{\partial \bar{\sigma}(w)}{\partial w_n}\\ &\quad + \left(\sigma_n \beta_n - \bar{\beta}(w)\frac{\partial \bar{\sigma}(w)}{\partial w_n}\right)\frac{\partial \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta}\Big|_{\beta=\bar{\beta}(w)}, \qquad (5)\end{aligned}$$

respectively, where

$$\frac{\partial \bar{\sigma}(w)}{\partial w_n} = \frac{\sum_{k=1}^N w_k \mathrm{cov}(R_n, R_k)}{\bar{\sigma}(w)} \quad \textit{for} \quad n \in \{1, 2, \ldots, N\},$$

and

$$\begin{aligned}\frac{\partial \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta} &= -\frac{\partial F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta}\\ &\quad - \frac{1}{\pi \eta}\mathrm{Re}\int_0^\infty e^{(iu+\delta)F^{-1}_{\mathrm{stdNTS}}(\eta,\alpha,\theta,\beta)}\frac{\phi_{\mathrm{stdNTS}}(-u+i\delta, \alpha, \theta, \beta)}{(-u+i\delta)^2}\\ &\quad \times \left((\delta + iu)\frac{\partial F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta} + \psi(-u+i\delta, \alpha, \theta, \beta)\right)du,\end{aligned}$$

with

$$\begin{aligned}&\psi(z, \alpha, \theta, \beta)\\ &= -zi + \left(1 - \frac{iz\beta}{\theta} + \left(1 - \frac{\beta^2(2-\alpha)}{2\theta}\right)\frac{z^2}{2\theta}\right)^{\frac{\alpha}{2}-1}\left(zi + \frac{\beta(2-\alpha)}{2\theta}z^2\right).\end{aligned}$$

5.1 Empirical illustration

In this subsection, we calculate MCT-VaR and MCT-CVaR of the NTS market model with the parameters fit to the empirical data and compare them to those values of a benchmark model. As a benchmark model, we consider the Gaussian market which is defined as

$$R = \mu + \mathrm{diag}(\sigma)Z, \quad Z \sim \Phi(0, \Sigma_Z),$$

where $\Phi(0, \Sigma_Z)$ is the N-dimensional standard normal distribution with covariance matrix Σ_Z whose diagonal entries are 1. The portfolio return in this model is

$$R_P = w^{\mathrm{T}} R \overset{\mathrm{d}}{=} \bar{\mu}(w) + \bar{\sigma}(w)\epsilon,$$

where $\epsilon \sim \Phi(0, 1)$, $\bar{\mu}(w) = w^{\mathrm{T}}\mu$, $\bar{\sigma}(w) = \sqrt{w^{\mathrm{T}}\mathrm{diag}(\sigma)\Sigma_Z\mathrm{diag}(\sigma)w}$ and the capital allocation weight vector w. Under this model, MCT-VaR and MCT-CVaR for tail level η is equal to

$$\frac{\partial \mathrm{VaR}_\eta(R_P(w))}{\partial w_n} = -\mu_n - F^{-1}_{\Phi(0,1)}(\eta)\frac{\sum_{k=1}^N w_k \mathrm{cov}(R_n, R_k)}{\bar{\sigma}(w)}, \tag{6}$$

and

$$\frac{\partial \mathrm{CVaR}_\eta(R_p(w))}{\partial w_n} = -\mu_n + \mathrm{CVaR}_\eta(\epsilon)\frac{\sum_{k=1}^N w_k \mathrm{cov}(R_n, R_k)}{\bar{\sigma}(w)}, \tag{7}$$

where $F^{-1}_{\Phi(0,1)}$ is the inverse function of the CDF of $\Phi(0, 1)$. Comparing MCT-VaR of (6) in the Gaussian model and MCT-VaR of (4) in the NTS model, (4) has one more term related to β, than (6). The asymmetry parameter β determines skewness, and the left/right tail decaying of the distribution. We can say the same argument for MCT-CVaR (5) and (7).

We consider an equally weighted portfolio with 30 stocks in Table 1, and use the estimated parameters in Table 2. The MCT-VaR and MCT-CVaR of the Gaussian market model are calculated by (6) and (7) for $\eta = 0.01$. The MCT-VaR and MCT-CVaR of the NTS market model are calculated by Proposition 2 and results are presented in Table 4. The table exhibits MCT-VaR and MCT-CVaR for the Gaussian model and the NTS model, and the rank of the values for the ascending order. The NTS market model describes skewness and fat-tails of portfolio return distribution. In contrast, the Gaussian market model is symmetric, and only the dispersion factor affects its VaR and CVaR. That generates different results between the Gaussian model column and the NTS model column in Table 4. For example, MCT-VaR and MCT-CVaR values for JPM rank 21st in the Gaussian model, while those values rank 14th in the NTS model. The MCT-VaR and MCT-CVaR values of WBA rank 19th and 17th, respectively, in the Gaussian model, while they rank 23rd and 24th, respectively, in the NTS model.

5.2 Risk budgeting

The MCT-VaR and MCT-CVaR are useful measures for portfolio managers to rebalance their portfolio allocation weight. That is, managers can reduce portfolio risk by decreasing their allocation weight to high-risk valued assets and increasing their allocation weight to low-risk valued assets. The distinction between high and low-risk assets is based on their MCT-VaR and MTC-CVaR.

Suppose a capital allocation weight vector w is given and let $\Delta w = (\Delta w_1, \Delta w_2, \ldots, \Delta w_N)^{\mathrm{T}} \in D$ where D is a zero neighborhood in $\mathbb{R}^N$ for a positive integer N. The optimal portfolios with respect to VaR and CVaR are obtained by solving the following problem:

$$\begin{aligned} &\min_{\Delta w} \Delta \mathrm{VaR}_\eta(R_P(w)) \\ &\text{subject to } \Delta E[R_P(w)] \geq 0 \text{ and } \sum_{n=1}^N \Delta w_n = 0, \end{aligned} \tag{8}$$

Table 4 MCT-VaR and MCT-CVaR for 30 stocks, under the Gaussian model and the NTS model

Symbol	Gaussian model				NTS Model			
	MCT-VaR (%)	Rank	MCT-CVaR (%)	Rank	MCT-VaR (%)	Rank	MCT-CVaR (%)	Rank
AAPL	2.23	24	2.58	24	2.91	22	3.99	22
AXP	1.97	20	2.27	20	2.59	17	3.54	17
BA	2.21	23	2.55	23	2.76	20	3.75	20
CAT	2.77	30	3.18	30	3.59	30	4.89	30
CSCO	2.36	26	2.71	26	3.27	29	4.49	29
CVX	1.70	12	1.96	12	2.27	13	3.10	13
DD	2.55	29	2.92	29	3.15	28	4.24	27
DIS	1.53	10	1.76	10	1.85	7	2.50	7
GS	2.38	27	2.73	27	3.00	24	4.05	23
HD	1.73	14	1.99	14	2.38	15	3.28	15
IBM	1.92	15	2.20	15	2.63	18	3.59	18
INTC	2.43	28	2.79	28	3.09	26	4.20	26
JNJ	1.34	7	1.54	7	1.88	8	2.60	8
JPM	1.97	21	2.27	21	2.35	14	3.17	14
KO	0.87	1	1.00	1	1.15	1	1.58	1
MCD	0.96	3	1.11	3	1.30	3	1.80	3
MMM	2.20	22	2.52	22	3.11	27	4.27	28
MRK	1.25	6	1.45	5	1.60	5	2.18	5
MSFT	2.27	25	2.62	25	3.01	25	4.13	25
NKE	1.96	17	2.25	18	2.44	16	3.32	16
PFE	1.45	8	1.67	9	2.02	9	2.77	9
PG	0.95	2	1.10	2	1.21	2	1.65	2

Table 4 continued

Symbol	Gaussian model				NTS Model			
	MCT-VaR (%)	Rank	MCT-CVaR (%)	Rank	MCT-VaR (%)	Rank	MCT-CVaR (%)	Rank
RTX	1.95	16	2.24	16	2.67	19	3.66	19
TRV	1.45	9	1.67	8	2.11	11	2.90	11
UNH	1.62	11	1.87	11	2.08	10	2.85	10
V	1.96	18	2.26	19	2.78	21	3.86	21
VZ	1.00	4	1.15	4	1.37	4	1.88	4
WBA	1.96	19	2.25	17	2.95	23	4.07	24
WMT	1.25	5	1.45	6	1.63	6	2.24	6
XOM	1.72	13	1.97	13	2.26	12	3.06	12

and

$$\min_{\Delta w} \Delta\mathrm{CVaR}_\eta(R_P(w))$$
$$\text{subject to } \Delta E[R_P(w)] \geq 0 \text{ and } \sum_{n=1}^{N} \Delta w_n = 0. \tag{9}$$

where

$$\Delta E[R_P(w)] = E[R_p(w+\Delta w)] - E[R_P(w)]$$
$$\Delta\mathrm{VaR}_\eta(R_p(w)) = \mathrm{VaR}_\eta(R_p(w+\Delta w)) - \mathrm{VaR}_\eta(R_p(w))$$
$$\Delta\mathrm{CVaR}_\eta(R_p(w)) = \mathrm{CVaR}_\eta(R_p(w+\Delta w)) - \mathrm{CVaR}_\eta(R_p(w)).$$

Since we have

$$\Delta\mathrm{VaR}_\eta(R_p(w)) \approx \sum_{n=1}^{N} \frac{\partial \mathrm{VaR}_\eta(R_p(w))}{\partial w_n} \Delta w_n,$$
$$\Delta\mathrm{CVaR}_\eta(R_p(w)) \approx \sum_{n=1}^{N} \frac{\partial \mathrm{CVaR}_\eta(R_p(w))}{\partial w_n} \Delta w_n$$

and $\Delta E[R_P(w)] = \mu^{\mathrm{T}} \Delta w$,

we can find the optimal portfolio on the local domain D with respect to VaR and CVaR, respectively, as follows:

$$\Delta w^* = \arg\min_{\Delta w} \sum_{n=1}^{N} \frac{\partial \mathrm{VaR}_\eta(R_p(w))}{\partial w_n} \Delta w_n$$
$$\text{subject to } \mu^{\mathrm{T}} \Delta w \geq 0 \text{ and } \sum_{n=1}^{N} \Delta w_n = 0. \tag{10}$$

and

$$\Delta w^* = \arg\min_{\Delta w} \sum_{n=1}^{N} \frac{\partial \mathrm{CVaR}_\eta(R_p(w))}{\partial w_n} \Delta w_n$$
$$\text{subject to } \mu^{\mathrm{T}} \Delta w \geq 0 \text{ and } \sum_{n=1}^{N} \Delta w_n = 0. \tag{11}$$

We perform the risk budgeting for VaR and CVaR using the 30 stocks in Table 1 with the estimated parameters in Table 2. Let the local domain be

$$D = \{(x_1, x_2, \ldots, x_{30}) \,|\, x_j \in [-2.5 \cdot 10^{-4}, 2.5 \cdot 10^{-4}],\ j = 1, 2, \ldots, 30\},$$

and the initial portfolio w_0 be the equally weighted portfolio. For risk budgeting for VaR under the NTS market model, we find Δw^* by (10) with (4). For risk budgeting for CVaR, we find Δw^* by (11) with (5). After that we put $w_{NTS}^{new} = w_0 + \Delta w^*$. We perform the risk budgeting under the Gaussian market model too for the benchmark. For risk budgeting for VaR, we find Δw^* by (10) with (6). For risk budgeting for CVaR, we find Δw^* by (11) with (5) under the NTS market model. After that we put $w_{Gauss}^{new} = w_0 + \Delta w^*$.

Table 5 provides the results of this optimization. To compare the portfolio's performance, we show the historical VaR and CVaR at 1% tail level. Since the historical VaR and CVaR are

Table 5 Risk budgeting result

	Initial historical risk (%)	Gaussian model (%)	NTS model (%)
$\text{VaR}_{1\%}(R_P(w))$	2.6077	2.6061	2.6056
$\Delta\text{VaR}_{1\%}(R_P(w))$		− 0.0594	− 0.0688
$\text{CVaR}_{1\%}(R_P(w))$	3.3109	3.3107	3.3106
$\Delta\text{CVaR}_{1\%}(R_P(w))$		− 0.0787	− 0.0950

VaR and CVaR in this table are historical VaR and historical CVaR

model free, we fairly compare the risk budgeting performance of the Gaussian market model to that of the NTS market model. In VaR and CVaR, both the Gaussian and NTS models have negative increments after risk budgeting, but the ΔVaR and ΔCVaR in the NTS model are smaller than those values in the Gaussian model, respectively. Hence, the risk budgeting of the NTS model performs better than that of the Gaussian model.

Next, we perform the risk budgeting for VaR and CVaR, iteratively as the following algorithm:

Step 1. Consider the equally weighted portfolio w as the initial portfolio.
Step 2. Calculate MCT-VaR or MCT-CVaR for w

- In risk budgeting for VaR, we use (4) for the NTS market model or use (6) for the Gaussian Model.
- In risk budgeting for CVaR, we use (5) for the NTS market model or use (7) for the Gaussian Model.

Step 3. Perform risk budgeting and find Δw^*

- In risk budgeting for VaR, we use (10).
- In risk budgeting for CVaR, we use (11).

Step 4. Change w to $w + \Delta w^*$ and go to Step 2. Repeat [Step 2 - Step 4] M times.

Using the 30 stocks in Table 1 with the estimated parameters in Table 2, we perform the iterative risk budgeting $M = 50$ times for

$$D = \{(x_1, x_2, \ldots, x_{30}) | x_j \in \left[-2.5 \cdot 10^{-4}, 2.5 \cdot 10^{-4}\right], j = 1, 2, \ldots, 30\}.$$

The results are exhibited in Fig. 8. For each iteration, we calculate historical VaR, and CVaR. Those values are drawn in the first and the second plates, respectively. The figure shows that

- VaR decreases in both the NTS Market model and the Gaussian market model, but the decreasing speed of the NTS model is faster than that of the Gaussian market model.
- CVaR decreases in both the NTS Market model and the Gaussian market model, but the decreasing speed of the NTS model is faster than that of the Gaussian market model.
- At iteration number 18, the slopes of the VaR and CVaR values are changed in the Gaussian market model case. Since the joint distribution of the empirical portfolio returns is not Gaussian, MCT-VaR and MCT-CVaR on the Gaussian model do not fit the empirical marginal contributions, and hence historical VaR and CVaR values do not decrease smoothly through the risk-budgeting. While, the NTS model fits the empirical distribution, and hence the risk budgeting makes the historical VaR and CVaR decrease smoothly.

Consequently, using risk budgeting of the NTS market model, we obtain the portfolio better performed than the portfolio obtained by the risk budgeting of the Gaussian market model.

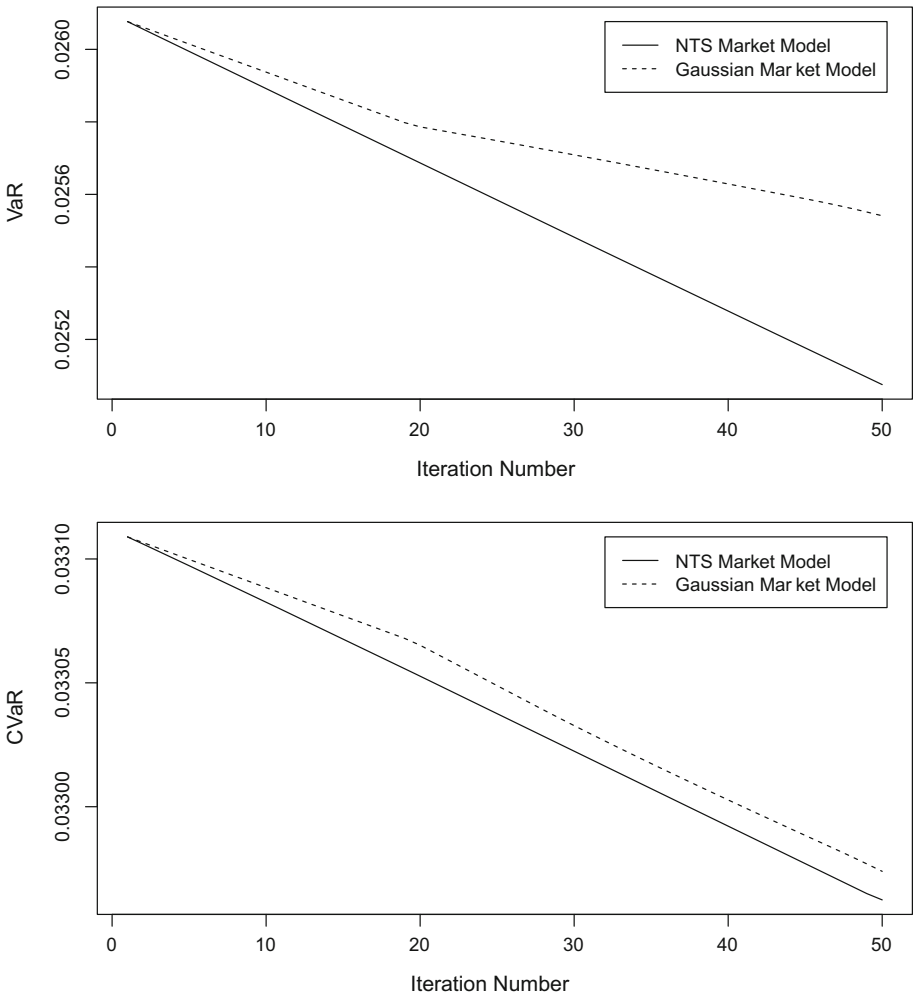


Fig. 8 Performance of risk budgeting. The first plate exhibit the VaR values for each iteration and the second plate exhibit CVaR values for each number of iteration. The solid lines are for the NTS market model and the dashed lines are for the Gaussian market model

6 Conclusion

In this paper, we discuss a generalized portfolio optimization considering not only mean and variance (dispersion) but also asymmetric tail risk. First, we constructed the NTS market model with the multivariate standard NTS noise. In Proposition 1, we find that parameters of portfolio return distribution are obtained by the marginal parameters and the covariance matrix of the portfolio returns. Using the NTS market model, we can decompose the portfolio risk to the dispersion risk and asymmetric tail risk, and obtain the generalized portfolio optimization method. The classical efficient frontier for mean-variance optimization is extended to a surface on the three-dimensional space by the NTS market model. The Sharpe ratio and tangency portfolio can be extended to the AS ratio and the AS ratio maximization portfolio.

Analytically tractable forms also give the marginal contribution to risk on the NTS model for VaR and CVaR.

Empirical tests are provided in the paper as well. The empirical surface of the efficient frontier on the NTS market model is presented for 30 major U.S. stock data. We show an empirical example that the AS ratio maximization strategy performs better than the Sharpe ratio maximizing strategy. Portfolio optimization methods by risk budgeting with the marginal contribution to VaR and CVaR are also performed on the NTS market model and the Gaussian market model. Using those empirical tests, we can see that the new portfolio optimization is more flexible and realistic than the traditional mean-variance method on the Gaussian model. The risk budgeting on the NTS market model increases the performance of the portfolio faster than the risk budgeting on the Gaussian market model.

A Appendix: Multivariate normal tempered stable distribution

Let $\alpha \in (0, 2)$ and $\theta > 0$, and let $\mathcal{T}$ be a positive random variable whose characteristic function $\phi_{\mathcal{T}}$ is equal to

$$\phi_{\mathcal{T}}(u) = \exp\left(-\frac{2\theta^{1-\frac{\alpha}{2}}}{\alpha}\left((\theta - iu)^{\frac{\alpha}{2}} - \theta^{\frac{\alpha}{2}}\right)\right). \tag{12}$$

The random variable $\mathcal{T}$ is referred to as *Tempered Stable Subordinator*. Let $X = (X_1, X_2, \cdots, X_N)^{\mathrm{T}}$ be a multivariate random variable given by

$$X = \mu + \beta(\mathcal{T} - 1) + \mathrm{diag}(\gamma)\varepsilon\sqrt{\mathcal{T}},$$

where

- $\mu = (\mu_1, \mu_2, \cdots, \mu_N)^{\mathrm{T}} \in \mathbb{R}^N$
- $\beta = (\beta_1, \beta_2, \cdots, \beta_N)^{\mathrm{T}} \in \mathbb{R}^N$
- $\gamma = (\gamma_1, \gamma_2, \cdots, \gamma_N)^{\mathrm{T}} \in \mathbb{R}_+^N$ with $\mathbb{R}_+ = [0, \infty)$
- $\varepsilon = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_N)^{\mathrm{T}}$ is a N-dimensional standard normal distribution with a covariance matrix Σ. That is, $\varepsilon_n \sim \Phi(0, 1)$ for $n \in \{1, 2, \ldots, N\}$ and (k, l)th element of Σ is given by $\rho_{k,l} = \mathrm{cov}(\varepsilon_k, \varepsilon_l)$ for $k, l \in \{1, 2, \ldots, N\}$.
- $\mathcal{T}$ is the Tempered Stable Subordinator with parameters (α, θ), and is independent of ε_n for all $n = 1, 2, \ldots, N$.

Then X is referred to as the *N-dimensional NTS random variable* with parameters $(\alpha, \theta, \beta, \gamma, \mu, \Sigma)$ which we denote by $X \sim \mathrm{NTS}_N(\alpha, \theta, \beta, \gamma, \mu, \Sigma)$. The NTS distribution has the following properties:

1. The mean of X are equal to $E[X] = \mu$.
2. The covariance between X_k and X_l is given by

$$\mathrm{cov}(X_k, X_l) = \rho_{k,l}\gamma_k\gamma_l + \beta_k\beta_l\left(\frac{2-\alpha}{2\theta}\right) \tag{13}$$

   for $k, l \in \{1, 2, \ldots, N\}$.
3. The variance of X_n is

$$\mathrm{var}(X_n) = \gamma_n^2 + \beta_n^2\left(\frac{2-\alpha}{2\theta}\right) \text{ for } n \in \{1, 2, \ldots, N\}.$$

4. Characteristic function of X_n is

$$\phi_{X_n}(u) = \exp\left((\mu-\beta)ui - \frac{2\theta^{1-\frac{\alpha}{2}}}{\alpha}\left(\left(\theta - i\beta u + \frac{\gamma^2 u^2}{2}\right)^{\frac{\alpha}{2}} - \theta^{\frac{\alpha}{2}}\right)\right)$$

Providing $\mu_n = 0$ and $\gamma_n = \sqrt{1-\beta_n^2\left(\frac{2-\alpha}{2\theta}\right)}$ with $|\beta_n| < \sqrt{\frac{2\theta}{2-\alpha}}$ for $n \in \{1,2,\cdots,N\}$, the N-dimensional NTS random variable X has $E[X] = (0,0,\ldots,0)^{\mathrm{T}}$ and $\mathrm{var}(X) = (1,1,\cdots,1)^{\mathrm{T}}$. In this case, X is referred to as the *N-dimensional standard NTS random variable* with parameters $(\alpha, \theta, \beta, \Sigma)$ and we denote it by $X \sim \mathrm{stdNTS}_N(\alpha, \theta, \beta, \Sigma)$.

For one dimensional NTS distribution, $\Sigma = 1$, we can prove the following Lemma which is changing parameterization.

Lemma 1 *Let $X \sim NTS_1(\alpha, \theta, \beta, \gamma, \mu, 1)$ and $\xi \sim \mathrm{stdNTS}_1(\bar{\alpha}, \bar{\theta}, \bar{\beta}, 1)$. Suppose $\bar{\alpha} = \alpha$, $\bar{\theta} = \theta$, and $\bar{\beta} = \beta/\sigma$, where $\sigma = \sqrt{\gamma^2 + \beta^2\left(\frac{2-\alpha}{2\theta}\right)}$. Then we have $X = \mu + \sigma\xi$.*

Proof The Ch.F of X is given by

$$\phi_X(u) = \exp\left((\mu-\beta)iu - \frac{2\theta^{1-\frac{\alpha}{2}}}{\alpha}\left(\left(\theta - i\beta u + \frac{\gamma^2 u^2}{2}\right)^{\frac{\alpha}{2}} - \theta^{\frac{\alpha}{2}}\right)\right) \tag{14}$$

By the definition of stdNTS distribution, the Ch.F of $\mu + \sigma\xi$ is equal to

$$\begin{aligned}\phi_{\mu+\sigma\xi}(u) &= E[e^{(\mu+\sigma\xi)ui}] = e^{\mu ui}E[e^{iu\sigma\xi}] \\ &= \exp\left((\mu-\bar{\beta}\sigma)iu - \frac{2\bar{\theta}^{1-\frac{\bar{\alpha}}{2}}}{\bar{\alpha}}\left(\left(\bar{\theta} - i\bar{\beta}\sigma u + \left(1-\bar{\beta}^2\left(\frac{2-\bar{\alpha}}{2\bar{\theta}}\right)\right)\frac{\sigma^2 u^2}{2}\right)^{\frac{\bar{\alpha}}{2}} - \bar{\theta}^{\frac{\bar{\alpha}}{2}}\right)\right).\end{aligned} \tag{15}$$

Hence (14)=(15) if $\bar{\alpha} = \alpha$, $\bar{\theta} = \theta$, $\bar{\beta}\sigma = \beta$, and $\gamma^2 = \sigma^2\left(1-\bar{\beta}^2\left(\frac{2-\alpha}{2\theta}\right)\right)$. Since $\sigma = \beta/\bar{\beta}$, we have

$$\gamma^2 = \frac{\beta^2}{\bar{\beta}^2}\left(1-\bar{\beta}^2\left(\frac{2-\alpha}{2\theta}\right)\right),$$

or

$$\gamma^2\bar{\beta}^2 = \beta^2\left(1-\bar{\beta}^2\left(\frac{2-\alpha}{2\theta}\right)\right) = \beta^2 - \beta^2\bar{\beta}^2\left(\frac{2-\alpha}{2\theta}\right),$$

and hence

$$\bar{\beta}^2\left(\gamma^2 + \beta^2\left(\frac{2-\alpha}{2\theta}\right)\right) = \beta^2.$$

Therefore, we have

$$\bar{\beta} = \frac{\beta}{\sigma},$$

where

$$\sigma = \sqrt{\gamma^2 + \beta^2\left(\frac{2-\alpha}{2\theta}\right)}.$$

□

The linear combination of NTS member variables of the NTS vector is again NTS distributed as the following proposition.

Lemma 2 *Let* $w = (w_1, w_2, \cdots, w_N)^T \in \mathbb{R}^N$ *and* $X \sim \mathrm{NTS}_N(\alpha, \theta, \beta, \gamma, \mu, \Sigma)$. *Then* $w^T X \sim \mathrm{NTS}_1(\alpha, \theta, \bar{\beta}, \bar{\gamma}, \bar{\mu}, 1)$, *where*

$$\bar{\mu} = w^T \mu, \quad \bar{\beta} = w^T \beta \quad \text{and} \quad \bar{\gamma} = \sqrt{w^T \mathrm{diag}(\gamma) \Sigma \mathrm{diag}(\gamma) w}.$$

Proof Since we have

$$w^{\mathrm{T}} X = w^{\mathrm{T}} \mu + w^{\mathrm{T}} \beta (\mathcal{T} - 1) + w^{\mathrm{T}} \mathrm{diag}(\gamma) \epsilon \sqrt{\mathcal{T}},$$

and $w^{\mathrm{T}} \mathrm{diag}(\gamma) \epsilon \overset{\mathrm{d}}{=} \sqrt{w^{\mathrm{T}} \mathrm{diag}(\gamma) \Sigma \mathrm{diag}(\gamma) w} \epsilon_0$ with $\epsilon_0 \sim \Phi(0, 1)$, it is trivial. □

Remark 1 In Lemma 2, the first four moments of $Y = w^{\mathrm{T}} X$ are as follows [See Rachev et al. (2011) and Kim et al. (2021)]

- Mean: $E[Y] = \bar{\mu}$
- Variance: $\mathrm{var}(Y) = \bar{\gamma}^2 + \bar{\beta}^2 \left(\dfrac{2 - \alpha}{2\theta} \right)$
- skewness: $\mathrm{S}(Y) = \dfrac{\bar{\beta} \ (2 - \alpha) \ \left(6 \bar{\gamma}^2 \theta - \alpha \bar{\beta}^2 + 4 \bar{\beta}^2 \right)}{\sqrt{2\theta} \left(2 \bar{\gamma}^2 \theta - \alpha \bar{\beta}^2 + 2 \bar{\beta}^2 \right)^{3/2}}$
- Excess kurtosis:
$$\mathrm{K}(Y) = \frac{(2-\alpha) \ \left(\alpha^2 \bar{\beta}^4 - 10 \alpha \bar{\beta}^4 - 12 \alpha \bar{\beta}^2 \bar{\gamma}^2 \theta + 24 \beta^4 + 48 \bar{\beta}^2 \bar{\gamma}^2 \theta + 12 \bar{\gamma}^4 \theta^2 \right)}{2\theta \left(2 \bar{\gamma}^2 \theta - \alpha \bar{\beta}^2 + 2 \bar{\beta}^2 \right)^2}$$

In general, higher moments can be obtained by the cumulants $c_n(Y)$:

$$c_n(Y) = \frac{1}{i^n} \left[\frac{d^n}{du^n} \log(\phi_Y(u)) \right]_{u=0}, n = 1, 2, \cdots.$$

Finally we can provide the proof of Proposition 1.

A.1 Proof of Proposition 1

Proof of Proposition 1 By (1) and (2), we have

$$\begin{aligned} R &= \mu + \mathrm{diag}(\sigma)(\beta(\mathcal{T} - 1) + \mathrm{diag}(\gamma) \epsilon \sqrt{\mathcal{T}}) \\ &= \mu + \mathrm{diag}(\sigma) \beta (\mathcal{T} - 1) + \mathrm{diag}(\sigma) \mathrm{diag}(\gamma) \epsilon \sqrt{\mathcal{T}} \\ &\sim \mathrm{NTS}_N(\alpha, \theta, \mathrm{diag}(\sigma)\beta, \mathrm{diag}(\sigma)\gamma, \mu, \Sigma), \end{aligned}$$

where $\gamma = (\gamma_1, \gamma_2, \cdots, \gamma_N)^{\mathrm{T}}$ with $\gamma_n = \sqrt{1 - \beta_n^2 \left(\frac{2-\alpha}{2\theta} \right)}$ and $\mathcal{T}$ is the tempered stable subordinator with parameter (α, θ). By Lemma 2,

$$R_P(w) = w^{\mathrm{T}} R \sim \mathrm{NTS}_1(\alpha, \theta, w^{\mathrm{T}} \mathrm{diag}(\sigma) \beta, \sqrt{w^{\mathrm{T}} \mathrm{diag}(\sigma) \mathrm{diag}(\gamma) \Sigma \mathrm{diag}(\gamma) \mathrm{diag}(\sigma) w}, w^{\mathrm{T}} \mu, 1).$$

By Lemma 1, we have

$$R_P(w) = w^{\mathrm{T}} \mu + \bar{\sigma}(w) \Xi$$

where

$$\bar{\sigma}(w) = \sqrt{w^{\mathrm{T}} \mathrm{diag}(\sigma) \mathrm{diag}(\gamma) \Sigma \mathrm{diag}(\gamma) \mathrm{diag}(\sigma) w + (w^{\mathrm{T}} \mathrm{diag}(\sigma) \beta)^2 \left(\frac{2 - \alpha}{2\theta} \right)}$$

and

$$\Xi \sim \text{stdNTS}_1(\alpha, \theta, w^{\mathrm{T}}\text{diag}(\sigma)\beta/\bar{\sigma}(w), 1).$$

Also, by (13), we have

$$\begin{aligned}(\bar{\sigma}(w))^2 &= \sum_{k=1}^{N}\sum_{l=1}^{N} w_k w_l \gamma_k \gamma_l \sigma_k \sigma_l \rho_{k,l} + \left(\sum_{k=1}^{N} w_k \sigma_k \beta_k\right)^2 \left(\frac{2-\alpha}{2\theta}\right) \\ &= \sum_{k=1}^{N}\sum_{l=1}^{N} w_k w_l \sigma_k \sigma_l \left(\gamma_k \gamma_l \rho_{k,l} + \beta_k \beta_l \left(\frac{2-\alpha}{2\theta}\right)\right) \\ &= \sum_{k=1}^{N}\sum_{l=1}^{N} w_k w_l \sigma_k \sigma_l \text{cov}(X_k, X_l).\end{aligned}$$

Hence, we have

$$(\bar{\sigma}(w))^2 = \sum_{k=1}^{N}\sum_{l=1}^{N} w_k w_l \text{cov}(R_k, R_l) = w^{\mathrm{T}} \Sigma_R w$$

where Σ_R is the covariance matrix of R. □

A.2 Proof of Proposition 2

Proof of Proposition 2 By (2), we have

$$R_P(w) = \bar{\mu}(w) + \bar{\sigma}(w)\Xi \quad \text{for} \quad \Xi \sim \text{stdNTS}_1(\alpha, \theta, \bar{\beta}(w), 1),$$

where

$$\begin{aligned}\bar{\mu}(w) &= w^{\mathrm{T}}\mu = \sum_{k=1}^{N} w_k \mu_k, \\ \bar{\beta}(w) &= \frac{w^{\mathrm{T}}\text{diag}(\sigma)\beta}{\bar{\sigma}(w)} = \frac{\sum_{k=1}^{N} w_k \sigma_k \beta_k}{\bar{\sigma}(w)}, \\ \bar{\sigma}(w) &= \sqrt{w^{\mathrm{T}}\Sigma_R w} = \sqrt{\sum_{k=1}^{N}\sum_{l=1}^{N} w_k w_l \text{cov}(R_k, R_l)}.\end{aligned}$$

Hence, the first derivative of $\bar{\beta}(w)$ and $\bar{\sigma}(w)$ are obtained as follows:

$$\begin{aligned}\frac{\partial \bar{\beta}(w)}{\partial w_n} &= \frac{\partial}{\partial w_n}\left(\frac{\sum_{k=1}^{N} w_k \sigma_k \beta_k}{\bar{\sigma}(w)}\right) \\ &= \frac{\sigma_n \beta_n \bar{\sigma}(w) - \left(\sum_{k=1}^{N} w_n \sigma_k \beta_k\right)\frac{\partial \bar{\sigma}(w)}{\partial w_n}}{(\bar{\sigma}(w))^2} \\ &= \frac{\sigma_n \beta_n}{\bar{\sigma}(w)} - \frac{\bar{\beta}(w)}{\bar{\sigma}(w)}\frac{\partial \bar{\sigma}(w)}{\partial w_n} \qquad (16)\end{aligned}$$

and

$$\frac{\partial \bar{\sigma}(w)}{\partial w_n} = \frac{1}{2\bar{\sigma}(w)} \frac{\partial}{\partial w_n} \sum_{k=1}^{N} \sum_{l=1}^{N} w_k w_l \mathrm{cov}(R_k, R_l)$$
$$= \frac{\sum_{k=1}^{N} w_k \mathrm{cov}(R_n, R_k)}{\bar{\sigma}(w)}.$$

Since we have

$$\mathrm{VaR}_\eta(R_p(w)) = -\bar{\mu}(w) - \bar{\sigma}(w) F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w))$$

we obtain

$$\begin{aligned}
&\frac{\partial \mathrm{VaR}_\eta(R_p(w))}{\partial w_n} \\
&= -\frac{\partial \bar{\mu}(w)}{\partial w_n} - \frac{\partial \bar{\sigma}(w)}{\partial w_n} F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w)) - \bar{\sigma}(w) \frac{\partial F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w))}{\partial w_n} \\
&= -\mu_n - \frac{\partial \bar{\sigma}(w)}{\partial w_n} F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \bar{\beta}(w)) + \bar{\sigma}(w) \frac{\partial F^{-1}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)}{\partial \beta}\Big|_{\beta=\bar{\beta}(w)} \frac{\partial \bar{\beta}(w)}{\partial w_n}
\end{aligned} \tag{17}$$

By substituting (16) into (17), we obtain (4).

Since there is $\delta > 0$ such that $|\phi_\Xi(-u + i\delta)| < \infty$ for all $u \in \mathbb{R}$, we have

$$\frac{\partial \mathrm{CVaR}_\eta(\Xi)}{\partial \beta}\Big|_{\beta=\bar{\beta}(w)} = \frac{\partial}{\partial \beta} \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta)\Big|_{\beta=\bar{\beta}(w)}.$$

By (3), we have

$$\begin{aligned}
&\frac{\partial}{\partial \beta} \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta) \\
&\quad = -\frac{\partial}{\partial \beta} F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta) \\
&\qquad - \frac{1}{\pi \eta} \mathrm{Re} \int_0^\infty e^{(iu+\delta) F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta)} \frac{\phi_{\mathrm{stdNTS}}(-u + i\delta, \alpha, \theta, \beta)}{(-u + i\delta)^2} \\
&\times \left((\delta + iu) \frac{\partial}{\partial \beta} F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta) + \frac{\partial}{\partial \beta} \log \phi_{\mathrm{stdNTS}}(-u + i\delta, \alpha, \theta, \beta) \right) du.
\end{aligned}$$

By setting $\psi(z, \alpha, \theta, \beta) = \frac{\partial}{\partial \beta} \log \phi_{\mathrm{stdNTS}}(z, \alpha, \theta, \beta)$, we can simplify

$$\begin{aligned}
\frac{\partial}{\partial \beta} \mathrm{CVaR}_{\mathrm{stdNTS}}(\eta, \alpha, \theta, \beta) &= -\frac{\partial}{\partial \beta} F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta) \\
&\quad - \frac{1}{\pi \eta} \mathrm{Re} \int_0^\infty e^{(iu+\delta) F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta)} \frac{\phi_{\mathrm{stdNTS}}(-u + i\delta, \alpha, \theta, \beta)}{(-u + i\delta)^2} \\
&\quad \times \left((\delta + iu) \frac{\partial}{\partial \beta} F^{-1}_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta) + \psi(-u + i\delta, \alpha, \theta, \beta) \right) du.
\end{aligned}$$

The characteristic function $\phi_{\mathrm{stdNTS}}(u, \alpha, \theta, \beta)$ is equal to

$$\phi_{\text{stdNTS}}(u,\alpha,\theta,\beta) = \exp\left(-\beta u i - \frac{2\theta^{1-\frac{\alpha}{2}}}{\alpha}\left(\left(\theta - i\beta u + \left(1-\frac{\beta^2(2-\alpha)}{2\theta}\right)\frac{u^2}{2}\right)^{\frac{\alpha}{2}} - \theta^{\frac{\alpha}{2}}\right)\right),$$

hence we have

$$\begin{aligned}\psi(z,\alpha,\theta,\beta) &= \frac{\partial}{\partial\beta}\left(-\beta z i - \frac{2\theta^{1-\frac{\alpha}{2}}}{\alpha}\left(\left(\theta - i\beta z + \left(1-\frac{\beta^2(2-\alpha)}{2\theta}\right)\frac{z^2}{2}\right)^{\frac{\alpha}{2}} - \theta^{\frac{\alpha}{2}}\right)\right)\\ &= -zi + \left(1 - \frac{iz\beta}{\theta} + \left(1-\frac{\beta^2(2-\alpha)}{2\theta}\right)\frac{z^2}{2\theta}\right)^{\frac{\alpha}{2}-1}\left(zi + \frac{\beta(2-\alpha)}{2\theta}z^2\right).\end{aligned}$$

As VaR case, CVaR for $R_P(w)$ is calculated using $\text{CVaR}\eta(\Xi)$ that

$$\text{CVaR}_\eta(R_p(w)) = -\bar{\mu}(w) + \bar{\sigma}(w)\text{CVaR}_\eta(\Xi).$$

Therefore, we have

$$\begin{aligned}\frac{\partial \text{CVaR}_\eta(R_p(w))}{\partial w_n} &= -\mu_n + \frac{\partial\bar{\sigma}(w)}{\partial w_n}\text{CVaR}_\eta(\Xi) + \bar{\sigma}(w)\frac{\partial \text{CVaR}_\eta(\Xi)}{\partial w_n}\\ &= -\mu_n + \frac{\partial\bar{\sigma}(w)}{\partial w_n}\text{CVaR}_\eta(\Xi) + \bar{\sigma}(w)\frac{\partial \text{CVaR}_\eta(\Xi)}{\partial\beta}\Big|_{\beta=\bar{\beta}(w)}\frac{\partial\bar{\beta}(w)}{\partial w_n},\end{aligned}$$

and, by (16), we obtain

$$\begin{aligned}&\frac{\partial \text{CVaR}_\eta(R_p(w))}{\partial w_n}\\ &= -\mu_n + \frac{\partial\bar{\sigma}(w)}{\partial w_n}\text{CVaR}_\eta(\Xi) + \left(\sigma_n\beta_n - \bar{\beta}(w)\frac{\partial\bar{\sigma}(w)}{\partial w_n}\right)\frac{\partial \text{CVaR}_\eta(\Xi)}{\partial\beta}\Big|_{\beta=\bar{\beta}(w)}\end{aligned} \tag{18}$$

By substituting $\text{CVaR}_\eta(\Xi) = \text{CVaR}_{\text{stdNTS}}(\eta,\alpha,\theta,\bar{\beta}(w))$ into (18), we obtain (5). □

References

Aas, K., Hobæk Haff, I., & Dimakos, X. K. (2006). Risk estimation using the multivariate normal inverse Gaussian distribution. *Journal of Risk, 8*(2), 39–60.

Adcock, C. (2010). Asset pricing and portfolio selection based on the multivariate extended skew-Student-*t* distribution. *Annals of Operations Research, 176*, 221–234.

Anand, A., Li, T., Kurosaki, T., & Kim, Y. S. (2016). Foster–Hart optimal portfolios. *Journal of Banking and Finance, 68*, 117–130.

Anand, A., Li, T., Kurosaki, T., & Kim, Y. S. (2017). The equity risk posed by the too-big-to-fail banks: A foster-hart estimation. *Annals of Operations Research, 253*(1), 21–41.

Barndorff-Nielsen, O. E., & Levendorskii, S. (2001). Feller processes of normal inverse Gaussian type. *Quantitative Finance, 1*, 318–331.

Barndorff-Nielsen, O. E., & Shephard, N. (2001). Normal modified stable processes. Economics Series Working Papers from University of Oxford, Department of Economics 72.

Bianchi, M. L., & Tassinari, G. L. (2020). Forward-looking portfolio selection with multivariate non-Gaussian models. *Quantitative Finance, 20*(10), 1645–1661.

Bianchi, M. L., Tassinari, G. L., & Fabozzi, F. J. (2016). Riding with the four horsemen and the multivariate normal tempered stable model. *International Journal of Theoretical and Applied Finance, 19*(4), 1650027.

Boyarchenko, S. I., & Levendorskiĭ, S. Z. (2000). Option pricing for truncated Lévy processes. *International Journal of Theoretical and Applied Finance, 3*, 549–552.

Carr, P., Geman, H., Madan, D., & Yor, M. (2002). The fine structure of asset returns: An empirical investigation. *Journal of Business, 75*(2), 305–332.
Carr, P., Geman, H., Madan, D., & Yor, M. (2003). Stochastic volatility for Lévy processes. *Mathematical Finance, 13*, 345–382.
Dahlquist, M., Farago, A., & Tédongap, R. (2017). Asymmetries and portfolio choice. *The Review of Financial Studies, 30*(2), 667–702.
Eberlein, E., & Glau, K. (2014). Variational solutions of the pricing pides for European options in Lévy models. *Applied Mathematical Finance, 21*(5), 417–450.
Eberlein, E., Gehrig, T., & Madan, D. (2012). Pricing to acceptability: With applications to valuation of one's own credit risk. *Journal of Risk, 15*, 91–120. https://doi.org/10.21314/JOR.2012.252
Eberlein, E., & Madan, D. B. (2010). On correlating Lévy processes. *Journal of Risk, 13*(1), 3–16.
Eberlein, E., & Özkan. (2005). The Lévy LIBOR model. *Finance and Stochastics, 9*, 327–348.
Fallahgoul, H. A., Kim, Y. S., Fabozzi, F. J., & Park, J. (2019). Quanto option pricing with Lévy models. *Computational Economics, 53*(3), 1279–1308.
Fama, E. (1963). Mandelbrot and the stable Paretian hypothesis. *Journal of Business, 36*, 420–429.
Gil-Pelaez, J. (1951). Note on the inversion theorem. *Biometrika, 38*, 3–4.
Gourieroux, C., Laurent, J., & Scaillet, O. (2000). Sensitivity analysis of values at risk. *Journal of Empirical Finance, 7*, 225–245.
Harvey, C. R., Liechty, J. C., Liechty, M. W., & Müller, P. (2010). Portfolio selection with higher moments. *Quantitative Finance, 10*(5), 469–485. https://doi.org/10.1080/14697681003756877
Hitaj, A., Mercuri, L., & Rroji, E. (2019). Sensitivity analysis of mixed tempered stable parameters with implications in portfolio optimization. *Computational Management Science, 16*, 71–95. https://doi.org/10.1080/00949655.2021.1962878
Kim, S. I., & Kim, Y. S. (2018). Normal tempered stable structural model. *Review of Derivatives Research, 21*(1), 119–148.
Kim, Y., Lee, J., Mittnik, S., & Park, J. (2015). Quanto option pricing in the presence of fat tails and asymmetric dependence. *Journal of Econometrics, 187*(2), 512–520.
Kim, Y. S., Giacometti, R., Rachev, S. T., Fabozzi, F. J., & Mignacca, D. (2012). Measuring financial risk and portfolio optimization with a non-Gaussian multivariate model. *Annals of Operations Research, 201*(1), 325–343.
Kim, Y. S., Rachev, S. T., Bianchi, M. L., & Fabozzi, F. J. (2010). Computing VaR and AVaR in infinitely divisible distributions. *Probability and Mathematical Statistics, 30*(2), 223–245.
Kim, Y. S., Roh, K. H., & Douady, R. (2021). Tempered stable processes with time-varying exponential tails. *Quantitative Finance*.https://doi.org/10.1080/14697688.2021.1962958, to appear.
Kim, Y. S., & Volkmann, D. (2013). NTS copula and finance. *Applied Mathematics Letters, 26*, 676–680.
King, A. J. (1993). Asymmetric risk measures and tracking models for portfolio optimization under uncertainty. *Annals of Operations Research, 45*, 165–177.
Konno, H., Shirakawa, H., & Yamazaki, H. (1993). A mean-absolute deviation-skewness portfolio optimization model. *Annals of Operations Research, 45*, 205–220.
Koponen, I. (1995). Analytic approach to the problem of convergence of truncated Lévy flights towards the Gaussian stochastic process. *Physical Review E, 52*, 1197–1199.
Kraus, A., & Litzenberger, R. (1976). Skewness preference and the valuation of risk assets. *Journal of Finance, 31*, 1085–1100.
Kurosaki, T., & Kim, Y. S. (2018). Foster–Hart optimization for currency portfolio. *Studies in Nonlinear Dynamics & Econometrics, 23*(2):Published Online. https://doi.org/10.1515/snde-2017-0119.
Liu, Y., Djurić, P. M., Kim, Y. S., Rachev, S. T., & Glimm, J. (2021). Systemic risk modeling with lévy copulas. *Journal of Risk and Financial Management*. https://doi.org/10.3390/jrfm14060251.
Mandelbrot, B. B. (1963a). New methods in statistical economics. *Journal of Political Economy, 71*, 421–440.
Mandelbrot, B. B. (1963b). The variation of certain speculative prices. *Journal of Business, 36*, 394–419.
Mansini, R., Ogryczak, W., & Speranza, M. G. (2007). Conditional value at risk and related linear programming models for portfolio optimization. *Annals of Operations Research, 152*, 227–256.
Markowitz, H. (1952). Portfolio selection. *Journal of Finance, 7*(1), 77–91.
Øigård, T. A., Hanssen, A., Hansen, R. E., & Godtliebsen, F. (2005). EM-estimation and modeling of heavy-tailed processes with the multivariate normal inverse Gaussian distribution. *Signal Processing, 85*, 1655–1673.
Pflug, G. (2000). Some remarks on the value-at-risk and the conditional value-at-risk. In S. Uryasev (Ed.), *Probabilistic constrained optimization: Methodology and applications* (pp. 272–281). The Netherlands: Kluwer Academic Publishers.
Rachev, S. T., & Mittnik, S. (2000). *Stable Paretian models in finance*. New York: Wiley.

Rachev, S. T., Stoyanov, S., & Fabozzi, F. J. N. J. (2007). *Advanced stochastic models, risk assessment, and portfolio optimization: The ideal risk, uncertainty, and performance measures*. Hoboken: Wiley.
Rachev, S. T., Kim, Y. S., Bianchi, M. L., & Fabozzi, F. J. (2011). *Financial models with Lévy processes and volatility clustering*. New York: Wiley.
Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk, 2*(3), 21–41.
Rockafellar, R. T., & Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance, 26*, 1443–1471.
Rubinstein, M. (1973). The fundamental theorem of parameter-preference security valuation. *Journal of Financial and Quantitative Analysis, 8*, 61–69.
Sharpe, W. F. (1966). Mutual funds performance. *Journal of Business, 36*(1), 119–138.
Sharpe, W. F. (1994). The sharpe ratio. *Journal of Portfolio Management, 21*(1), 45–58.
Stoyanov, S. V., Racheva-Iotova, B., Rachev, S. T., & Fabozzi, F. J. (2010). Stochastic models for risk estimation in volatile markets: A survey. *Annals of Operations Research, 176*(1), 293–309.
Stoyanov, S. V., Rachev, S. T., & Fabozzi, F. J. (2013). Sensitivity of portfolio VaR and CVaR to portfolio return characteristics. *Annals of Operations Research, 205*, 169–187.
Xia, Y., & Grabchak, M. (2021). Estimation and simulation for multivariate tempered stable distributions. *Journal of Statistical Computation and Simulation*. https://doi.org/10.1080/00949655.2021.1962878

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.